PipedPeer: Zero-Configuration Distributed Computing via Recursive, Decentralised Coordination

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***Abstract*—Distributed computing frameworks presuppose a configured cluster, a program written against their API, and machines that can reach one another. PipedPeer removes all three for Python: a single command runs an unmodified script on other machines, each running the same daemon, with no head node and no scheduler process. The script's environment is built as a Nix closure against a pinned package set, so it has the same store path on every machine, and a peer receives only the store paths it lacks. Placement runs in the submitting client under a lease, and every distributed path falls back to a local one. The parallel primitives the script already uses are substituted at import time; parallel maps are divided by a makespan solver over measured peer throughput, and numerical work moves only when a cost model predicts a gain. PyTorch data-parallel training is synchronised through the daemons with batches weighted by measured speed, either exactly or by DiLoCo-style periodic averaging; at the default two-rank interval of 32 steps, averaging stayed within 1% of one device's validation loss on every test model while each rank sent 7.6–9.2$\times$ fewer bytes than exact synchronisation. Machines behind different routers meet through an introducer that any member can serve, then connect directly over authenticated QUIC; no data is relayed. In a container lab a repeat submission took 0.83 s against 23.4 s for the first, and interception cost $1.24\times$ the startup of a trivial job. Between two laptops in different homes, over a Tailscale network, a CPU-bound script ran in about 14 s against 18 s on one.**

***Index Terms*—distributed computing, peer-to-peer, NAT traversal, transparent interception, reproducible environments, content-addressed caching, cost-based scheduling**

I. Introduction

The gap between running `python script.py` and running the same work on several machines is large in operational effort and often small in the computation involved. Apache Spark [1], Ray [2] and Dask [3] presuppose a configured cluster with a head node, matching software on every worker, a scheduler process, and a network on which every machine reaches every other. Remote execution over `ssh` avoids the setup but provides no environment management, placement, isolation or fault tolerance, and still needs a reachable address.

Three obstacles are usually treated separately. The *environment* obstacle: a remote machine must be made to match the local one. The *code* obstacle: using more than one machine normally means rewriting the program. The *reachability* obstacle: machines behind different home or campus routers accept no inbound connections, and the usual answers — a VPN, port forwarding or a relay — are either configuration or a bottleneck. PipedPeer addresses all three. Its contributions are:

1) **Peer-to-peer closure distribution.** Building on Nix's content-addressed store paths [4], we push signed closures between peers, send only the store paths a receiver lacks, and offer distributed work only to peers that can run the closure (Section III-B).
2) **Transparent interception with measured work sharing.** Parallel primitives are substituted at import time, parallel maps are divided by a makespan solver over measured rates, and numerical work is gated by a cost model (Sections III-D, III-E).
3) **Joining across networks without a relay**, through an introducer that only exchanges addresses and that every daemon serves (Section III-C).
4) **Placement without a scheduler**: filters and a bounded score evaluated in the submitting client, with a three-phase lease (Section III-A).
5) **A fallback that preserves the result on every distributed path** (Section III-F).
6) **Speed-weighted data-parallel training through the daemon**, with exact or fast (DiLoCo-style) synchronisation chosen by ring size (Sections III-D, IV-D).

TABLE I
COMPARISON WITH EXISTING SYSTEMS. THE LOWER BLOCK IS WHERE OURS IS PARTIAL OR THE WEAKER SYSTEM

Property	Ours	Spark	Ray	Dask	ssh
Runs unmodified Python	yes	no	part.	part.	yes
No cluster to configure	yes	no	no	no	part.
No head or master node	yes	no	no	no	yes
Reproduces the environment	yes	part.	part.	part.	no
Joins NATed machines, no relay†	yes	no	no	no	no
Survives losing a node*	part.	yes	yes	yes	no
Authenticated, encrypted	part.	yes	part.	part.	yes
Sandboxes each job	part.	no	no	no	no
Runtimes beyond Python	**no**	yes	part.	no	yes
Proven in production	**no**	yes	yes	yes	yes

†When one introducer is reachable; otherwise a public address or an overlay is needed.
*Any worker; not the submitter of a parallel script or the machine of a training ring's rank 0.

II. RELATED WORK

Spark [1], Ray [2] and Dask [3] are mature and far more capable at scale. Ray ships a drop-in `multiprocessing.Pool` and a `joblib` backend, and Modin [5] provides distributed pandas behind the pandas API; each needs a changed import and a running cluster. PipedPeer changes no line of the program, because substitution happens before the first user statement, and it decides per call whether to distribute. Table I summarises the differences. BOINC [6] pioneered volunteer computing, but as a platform for specific projects rather than a general executor.

The cross-network mechanism follows established practice. Reflexive addresses are discovered as in STUN [7]; port mappings are requested with PCP [8], NAT-PMP [9] and UPnP-IGD [10]; simultaneous UDP probing from the socket that will carry the connection is the hole punching of Ford et al. [11]; links are QUIC [12].

Edge-computing offloading studies the decision our cost model makes. Deng et al. [13] weigh transmission against local computation; Yong et al. [14] show partial offloading beating all-or-nothing, the regime our interception works in; learned policies [15] exceed a fixed rule at the price of training; and topology-aware placement [16] outperforms scalar scoring, a limitation our placement shares. Spenger [17] defines failure transparency, under which application code never observes a failure; Section III-F must provide it for scripts that contain no failure handling. For training we build on PyTorch DDP [18], 8-bit gradient compression [19] with error feedback [20], and periodic averaging as in DiLoCo [21] and local SGD [22].

III. SYSTEM DESIGN

Design goals. (G1) The script runs unchanged. (G2) Joining needs one setup command, plus one code across networks. (G3) No machine is special: there is no head node, scheduler or registry, and every machine both submits and works. (G4)

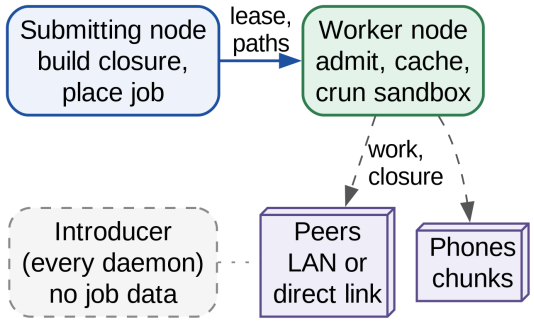


Fig. 1. System design. Both roles are the same binary; peers on other networks are reached over direct links that the introducer helps establish but never carries.

Distribution never changes the result and is declined when it would be slower. (G5) No job data passes through a third machine.

Coordination is *decentralised*: every daemon serves the introducer for its cluster, placement is computed in the submitting client, and a training ring's lead is simply the node that runs rank 0 of that job. It is *recursive* in that division repeats at each level: the node that holds a parallel map splits it among its peers, and each part is split again across the cores of the node that runs it. A node that cannot hold a map forwards it whole, once, to a peer that can, and that peer makes the split; a part already split off is never split across machines or forwarded again.

Every machine runs a single daemon; the command-line client is short-lived. The coordinator is a library linked into the client, so placement is decided where the job is submitted (Fig. 1). On one network participants find each other unaided; across networks a participant runs `pipedpeer connect` with a one-line code that another member's `pipedpeer invite` printed, carrying the cluster secret and a few introducer addresses.

A. Placement

Candidates come from known peers, LAN discovery and internet-linked peers. Hard filters run in order — architecture, GPU if required, then free memory, cores and VRAM — and survivors are scored:

$$S = \Big(1 - \frac{u_{\text{cpu}} + u_{\text{mem}}}{200} - 0.05\, n_{\text{jobs}} + 0.10(\hat{c} - 0.5) + 0.06(\hat{f} - 0.5) + 0.10(\hat{m} - 0.5) + 0.10(\hat{r} - 0.5)\Big) \cdot h \tag{1}$$

where u_{cpu} and u_{mem} are processor and memory utilisation in percent, n_{jobs} the running jobs, $\hat{c}, \hat{f}, \hat{m}, \hat{r}$ core count, clock, free memory and free-core ratio normalised onto $[0, 1]$, and $h \in [0, 1]$ the node's health score; S is clamped at zero, and a GPU request adds up to 0.8 for GPU capacity. Load dominates hardware: the utilisation terms can subtract 1.0, the hardware terms move the score by at most ± 0.18, and a missing metric

takes the neutral 0.5. Free memory is what a node can *grant*: where a user `systemd` is available, each daemon runs under a cgroup ceiling of half the machine, and co-located daemons (several on one machine) subtract one another's reservations.

Work is reserved by a three-phase lease; the concurrency and memory checks and the insertion happen under one lock. The submitter renews every 30 s, also while the environment builds and uploads, against a five-minute expiry. A submitter that dies stops renewing and its slot is reclaimed; a worker that dies leaves the healthy set and its job is rescheduled with back-off.

B. Environment Reproduction

The client scans imports, resolves them to packages, and builds a Nix closure from a flake pinned to an *exact* package-set revision, so the store path is a stable name for an environment across machines and time. Imports absent from the mapping table are looked up by name in the pinned set, the scan follows the script's own modules, a `requirements.txt` is honoured, and an import still unresolved is reported before the run.

Before uploading, the client asks whether the target can already run the closure; if not, it sends only the store paths the target lacks. After an upload the worker pushes the closure to peers that cannot yet run it, because only those that can are offered distributed work. Between daemons the push is a binary cache produced and imported by Nix itself and signed with the sending node's key; the receiver checks that key against the signer's published identity before trusting it for that import. Closures go only to peers of the same architecture; a peer of another architecture receives the package list and builds its own environment once. A phone joins under Termux [23], where `proot-distro` provides, through `proot` [24], a Debian root with single-user Nix, and takes parallel chunks only.

C. Joining Across Networks

An introducer keeps an address book, keyed by a hash of the cluster secret, and forwards punch requests between two nodes; it carries no job data. Every daemon serves one for its own cluster and registers with every introducer it knows, so no single machine's address is a point of failure. Each node offers its LAN addresses, a port mapped on its router (published only when the router's external address matches the one the introducer observed) and its reflexive address. Probes, introducer messages and QUIC share one UDP socket, because a router's mapping belongs to the socket that opened it; both sides probe every candidate pair every 250 ms for 15 s. When one side's NAT allocates a new port per destination, that side opens 256 sockets toward the peer while the other sends 600 probes to random ports, and the socket that is hit is kept [25], about 91% of rounds over 64,511 ports.

The winning path carries QUIC with a hello signed by each node's Ed25519 [26] key and checked against the node it meant to reach; the hello signs the session's exported keying material [27] and proves the cluster secret by an HMAC [28] over it, so it cannot be replayed on another session. The link appears locally as a loopback forwarder recorded as one path to that node, so placement, transfer and gradient exchange run over it unchanged. A node is identified by its key's fingerprint, never by its address: the node store keeps every path to it and switches only to a path that answered and proved the key, and QUIC connections migrate when an address changes, so a machine that changes network rejoins on its own. A peer with no path is reported with its reason and routed around, never relayed. The daemon API requires the cluster secret on every route and serves pinned TLS [29] and HTTP on a single port.

D. Transparent Interception

A Python package embedded in the binary is placed on the interpreter's path so that it is imported before the user's first line; libraries are patched on first import. Substituted primitives are parallel maps, `joblib`, pandas grouping, joining and chunked reading, NumPy matrix products and decompositions, and data-parallel training.

Parallel maps. A kernel travels as *source* when it is a module-level function, including one in `__main__`, or *by value* through cloudpickle [30] when it is a lambda or closure. The node that holds a map splits it once among its peers, and each peer runs its part on its own cores. The split is a makespan solver: each device is modelled as a setup time plus a rate, fitted from the parts it has returned, and a device whose setup alone exceeds the makespan is declined. A part unfinished past its prediction counts as taking at least that long, a part waiting or overdue is copied to a free peer, and the first copy to finish wins.

DataFrame operations. pandas grouping and joining hash rows by key, so each key is complete on a single node and any aggregation is exact with no combiner; the row order of joins is not preserved.

Training. Ranks post gradients to daemons, not to a rank-to-rank mesh: to the lead node's daemon, or, in exact mode on three or more directly reachable machines, one shard to each rank's daemon (a butterfly). Every rank receives one sample-weighted average $\sum_i n_i g_i / \sum_i n_i$, sent as 8-bit values with per-block scales and error feedback. The global batch is partitioned across ranks in proportion to their speed, preserving the effective learning rate. Unless the user fixed the ring size with `--ddp` N, after a warm-up the lead compares the split's step time with the fastest rank alone and keeps the ring only if it is faster. Samples are assigned to ranks by a weighted rendezvous hash [31]: when a rank dies the survivors take its share at the next epoch boundary while over 95% of their samples stay in place, and a rank that returns joins from the lead's model and optimiser snapshot, taking about $1/n$ of the samples.

A fast mode synchronises every H steps: an exact warm-up (a tenth of the run, 10–500 steps), then local steps whose weights are averaged by samples, with optimiser moments averaged every tenth round; H shrinks to 1 over the run's last tenth. It never changes epochs, batch or learning rate, and must

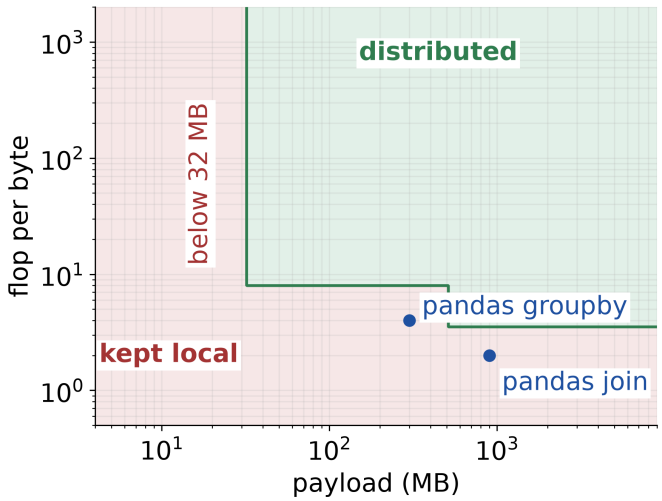


Fig. 2. Decision regions of Equation 2 for the pandas path ($\rho = 1$, 1 GFLOP/s) at three nodes on a 500 MB/s link. The step at 512 MiB is a separate rule: below 8 FLOP per byte, work up to 512 MiB stays local.

keep validation loss at most 1% above one device's (mean over seeds, every test model). DiLoCo's outer Nesterov step (lr 0.7, momentum 0.9) settles at $0.7/(1-0.9) = 7$ times the averaged change, which suits syncs every few hundred steps; at our intervals of 2–64 steps PipedPeer uses plain averaging (outer lr 1, no momentum), chooses H by ring size (Section IV-D), and uses no interval past four ranks. Under automatic sync the lead switches a ring to fast mode once, when the predicted fast step with a 10% margin beats the exact split and, for a ring not fixed by `--ddp` N, also the fastest rank alone; with `--ddp` N the ring size is fixed and auto chooses only the sync mode.

E. The Cost Model

Numerical work is gated. With b payload bytes, ϕ operations per byte, ρ link crossings, B bandwidth, R and R' local and remote throughput and K nodes, divisible work moves only when

$$\frac{b\phi}{R} > \frac{\rho\, b}{B} + \frac{b\phi}{R'K} \times 1.3. \tag{2}$$

Work needs at least two nodes and 32 MiB; work that does not fit in local memory moves regardless, and a matrix product that fits nowhere raises an error rather than exhausting the machine. A GPU peer raises R' sixfold, a conservative heuristic multiplier. The inequality is first tried at an optimistic 10 Gbit/s and, only if offloading could win, bandwidth is taken from recent transfers or an 8 MiB probe through the dispatch path (Fig. 2). Indivisible work (SVD) moves when transfer costs under half the local time. At about 200 GFLOP/s, local BLAS outruns a CPU cluster's transport, so dense matrix products stay local, while a GPU peer can win.

F. Degradation Guarantee and Operation

Using the cluster must never change the result. Parallel-map parts fill a result table first-wins, idle local cores re-run any part not yet returned, and the call returns once every item is covered, so an unresponsive cluster degrades to a local pool. Every intercepted call falls back to the original on exception, a failed peer is skipped, a dead worker is respawned, a failed job is rescheduled with back-off, and a training ring continues on the ranks that are left.

A run behaves like a local one. An interrupt sends one cancel frame; the executing daemon signals the job's process group, and parallel parts stop on every peer because each carries its request's context. A dropped connection never kills the job: the daemon logs every output frame, keeps the lease while the job runs, and lets the submitter reattach and replay the log without gaps or repeats. Each run ends with one line splitting its time into the script's and PipedPeer's, and a summary of where its items ran.

G. Implementation

PipedPeer is a single Go 1.25 binary that is both client and daemon, built from 46 Go packages and an embedded Python interception shim. The daemon serves an HTTP API, makes direct links with quic-go, builds environments with Nix flakes, and runs each job in a `crun` sandbox. A mutation audit breaks each of 95 features in turn; every mutation fails a test.

IV. Evaluation

We ask four questions. **Q1:** What does interception cost, including when it declines? **Q2:** How much do the environment cache and the closure diff save? **Q3:** Does work cross machines with correct results, and does it pay between physical machines? **Q4:** Does split training keep accuracy, and when is it faster?

A. Setup

A 12-thread Intel i5-13420H laptop with 15 GiB of memory under Fedora 43 runs a lab of three worker daemons in containers, each with a fixed processor and memory limit; where the network matters, links are shaped with `tc netem` to the latency, bandwidth and loss of a LAN, a Wi-Fi network or a 5G link. Because the containers share one processor, the lab measures correctness, overhead, caching and cost growth, and speedup only as one container against several of the same size. A 16-thread laptop with an RTX 3060 laptop GPU under Fedora 44 runs a second lab. Physical runs use two laptops in different homes (RTX 3060 and RTX 3050), a 20-core and a 16-core host joined over Tailscale, and a Samsung Galaxy S20 FE (Android 13). Between the two laptops, PipedPeer's own QUIC link ran on top of a Tailscale (WireGuard) network with 6–8 ms round trip, because both laptops joined one Tailscale network: neither home offered a reachable introducer, since one router refuses port mapping and the other machine sits behind two NATs. These runs measure distribution between homes, not PipedPeer's NAT traversal. NAT traversal without Tailscale is evaluated on a two-site container testbed on one host: two containers on separate private networks, both numbered 192.168.0.0/24, each behind a router container whose iptables rules translate

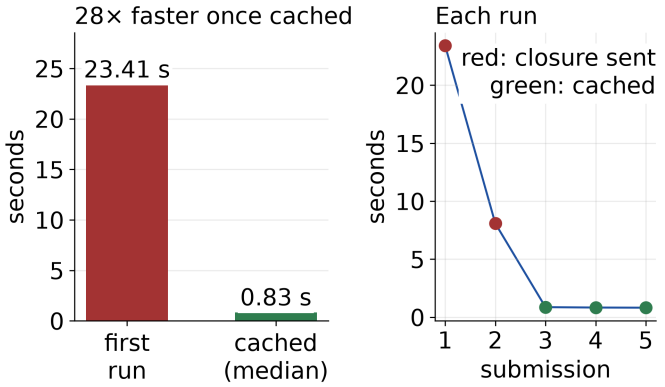


Fig. 3. Five submissions of the same environment in the container lab.

addresses and drop unsolicited inbound packets, with the introducer on a third network standing for the internet; every user command is run across it. Training accuracy is measured against one device trained on the same data: two-rank rings in the lab, larger rings in-process, with the ranks side by side on a single machine. Lab wall times are medians of three to five runs, and of one seed for training; lab accuracy is a mean of three seeds, in-process accuracy of seeds 0–2 and, where run, 0–5. MiB and GiB are binary.

B. Microbenchmarks (Q1, Q2)

Interception. Startup of a job that imports nothing heavy takes 0.117 s with interception against 0.094 s without (best of five, 1.24×). On a whole job that the cost model declines, a 16-item map with 0.01 s of work and none of it dispatched, the median was 0.80 s against 0.68 s: declining costs about 0.12 s per job.

Environment cache. The first submission of a job took 23.4 s. The second landed on a worker that had not yet received the closure and took 8.1 s; the next three found it in place, shipped only the workspace, and took a median of 0.83 s (Fig. 3). Adding one package to an environment a peer already held sent 29 MB instead of 206 MB.

Work split. On a lab limited to 8, 2 and 1 CPUs, a head-count split was 1.6× slower than local (3.6 s); the measured split declined the 1-CPU worker and took 2.9 s, better but still 1.3× slower than local. With one peer at a tenth of the CPU, the slowdown against an equal cluster fell from 1.6× on the first run to 1.08× by the fourth as the rate estimates converged.

Introducer. In a run between two physical machines the introducer's host carried 0.2 MB against about 30 MB between the peers, and with the introducer suspended a known peer was reached in 0.03 s.

C. End-to-end runs (Q3)

With distribution forced in the three-worker lab, over a third of a parallel map's items ran on the other daemons, with results identical to a local run. In the second lab a parallel map took 43.5 s on its first run and 17.0 s with its environment cached; a GPU job submitted from a node without a GPU ran on the GPU node in 7.6 s cached (264.8 s the first time); an interrupt stopped work on every peer within 2 s; after the introducer changed address, a three-node cluster reconnected with no command in about 12 s; a detached job was reattached and returned its results; and a two-rank ring finished its 80 epochs after one rank was killed, in 90.5 s.

TABLE II
RESULTS BETWEEN PHYSICAL MACHINES

Measurement	Result
Pool map, 20-core host alone vs. with 16-core peer (Tailscale)	5.8–6.4 vs. 4.9–5.4 s
CPU script, one laptop vs. two over Tailscale	18 vs. 14 s
Training, two laptops (Tailscale): 1 GPU / `--ddp 2` exact / fast / auto	4.2 / 37.1 / 8.5 / 14.6 s
val. loss: 1 GPU / exact / fast / auto	0.0794 / 0.0797 / 0.0797 / 0.0796
New GPU environment, first run vs. repeat	506 vs. 17.7 s

Table II lists the physical-machine results. A parallel map and a mixed CPU script (pool, joblib, NumPy, pandas) ran faster on two machines than on one (a job of about 5 s is dominated by setup and serialisation), the latter with about half its items on the second laptop. The first run of a new GPU environment is dominated by its build (487 s of 506 s, mostly downloading CUDA PyTorch); the same script then took 17.7 s. On the phone, a laptop's CPU script spread over both laptops and the phone gave the phone 8 of 192 items with an unchanged result, after the phone had built the environment from the cache in 27 s. Without its pandas step the script took a median of 62.8 s on the phone with Nix's Python in `proot`, against 73.1 s with Termux's (three runs each).

Across two NATs. On the two-site testbed (Section IV-A), scripts, parallel maps, remote runs and both training sync modes ran across the sites, and a node whose link was cut mid-run was replaced. Without a reachable introducer, a public address or an overlay such as Tailscale is needed.

D. Training (Q4)

When splitting pays. In the lab, a small model with heavy steps split over two containers beat one (9.1 against 11.8 s), while a 9 MB model with light steps over Wi-Fi ran ten times slower split than alone (80.7 against 8.0 s). Between the two laptops no split beat one GPU (Table II). Those rings were fixed with `--ddp 2`, so automatic sync could choose only the sync mode: the ring switched to fast mode after a short exact warm-up and finished in 14.6 s, 2.5× faster than exact; without `--ddp` N the lead releases a ring that one rank alone outruns. Offered 16-, 8- and 2-core ranks on one machine, admission took the first two, which ran in 23.6 s with measured batch sizes against 25.4 s with equal ones.

Fast synchronisation. At the default two-rank interval, plain averaging every 32 steps after a 10-step warm-up kept validation loss within 1% of one device in-process: +0.06% on a GPU-sized MLP (MLP-G), +0.64% on a CPU-sized

MLP (MLP-C) and $+0.73\%$ on a small transformer (seeds 0–2; $+0.35\%$ and -1.02% for the last two over seeds 0–5). In the lab at 32 steps (one seed), each rank sent 7.6–9.2$\times$ fewer bytes than exact sync, and the ring finished 1.2–1.4$\times$ faster than exact on LAN and 1.6–3.1$\times$ on Wi-Fi. At 64 steps the lab stayed within the bar too (three seeds, LAN and Wi-Fi, worst $+0.64\%$), but in-process the transformer failed on seeds 0–2 ($+2.13\%$). DiLoCo's outer step left the transformer 21.7% above one device even at one sync per epoch (156 local steps).

Many ranks. In-process runs, on up to six seeds, set the interval $H(n)$ as the largest that passed on every model: 32 at two ranks, 16 at three and 4 at four; at eight ranks the transformer failed at every interval down to 2, so no interval is used for five or more ranks. Of ten methods tested at 4, 8 and 11 ranks, only exact synchronisation with rank-16 PowerSGD [32] compression and error feedback kept every model within 1% (worst mean $+0.18\%$), with 3.2$\times$ fewer bytes than int8 on the MLPs but $1.7\times$ more on the transformer.

V. Limitations

The labs share one processor among their workers, so they show cost, not speedup, and the physical runs cover two or three machines with few repetitions. NAT traversal is evaluated on the two-site testbed (Section IV-A) and over an overlay between real homes. TLS is trust on first use, the keys trusted to sign closures are not yet bound to an authenticated identity, and authorisation is one shared secret. Only Python is supported, and a training ring depends on rank 0's machine. The 1% accuracy bar is a mean over seeds; single seeds spread from -3.9% to $+2.2\%$.

VI. Conclusion and Future Work

PipedPeer runs unmodified Python on other machines without configuring a cluster: it ships environments as content-addressed closures diffed by store path, substitutes the parallel primitives a script already uses, divides work by measured speed, and joins machines across networks over direct, authenticated links with no relay and no privileged node. Its central rule is to measure before distributing and to decline when that would be slower; the cost gate, the measured split and the ring's own check each follow from it. Future work is to bind signing keys to authenticated identities and require TLS, to add locality- and topology-aware placement and AMD accelerators, to let a training ring survive its submitter and rank 0, and to run a controlled evaluation on more machines.

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